

Endocrinology Clinical Assessment Report

Diabetes Mellitus Screening · Confidential Clinical Document

Patient Information

Patient Name	Sarah K.	Patient ID	DIA-2026-001
Date of Report	April 29, 2026	Exam Type	Endocrinology Screening — Diabetes Assessment
Age / Gender	42 years · Female	Referring Department	Internal Medicine
Pregnancies	6 (reported)	BMI	31.2 kg/m ² — Class I Obesity
Family History (DPF)	0.52 — Moderate hereditary risk	Informant	Patient / Medical Record

Laboratory & Clinical Findings

Parameter	Result	Reference Range	Status	Clinical Note
Glucose (glucose)	135 mg/dL	70–99 mg/dL	HIGH	Pre-diabetic to diabetic range — impaired fasting glucose
Blood Pressure (blood_pressure)	82 mmHg	60–80 mmHg (dia)	ELEVATED	Mildly elevated diastolic — hypertensive tendency
Skin Thickness (skin_thickness)	28 mm	10–25 mm	ELEVATED	Increased triceps skinfold — associated with insulin resistance
Insulin (insulin)	96 µU/mL	16–166 µU/mL	NORMAL	Within range but borderline — monitor for hyperinsulinemia
BMI (bmi)	31.2 kg/m ²	<25.0 kg/m ²	HIGH	Class I obesity — major independent risk factor for T2DM
Diabetes Pedigree (diabetes_pedigree_function)	0.52	0.0–0.4 (low risk)	MODERATE	Hereditary risk above threshold — significant family history
Age (age)	42 years	—	RISK	Age >40 elevates T2DM susceptibility significantly
Pregnancies (pregnancies)	6	<4 (lower risk)	ELEVATED	High parity — increased gestational diabetes history risk

6 out of 8 parameters deviate from normal reference ranges. Glucose, BMI, skin thickness, pedigree function, parity, and age all trend toward diabetic risk.

Model Input Features

pregnancies	6	insulin	96
glucose	135	bmi	31.2
blood_pressure	82	diabetes_pedigree_function	0.52
skin_thickness	28	age	42

These feature values are extracted directly from clinical findings and formatted for ingestion by the diabetes prediction model.

Clinical Observations

Sarah K. is a 42-year-old female presenting for diabetes mellitus screening. She reports a history of 6 pregnancies, raising concern for prior gestational diabetes episodes. Her fasting plasma glucose of 135 mg/dL falls within the diabetic range (≥ 126 mg/dL), indicating impaired glucose homeostasis. BMI is 31.2 kg/m², consistent with Class I obesity, a well-established risk factor for insulin resistance and Type 2 diabetes. Skin thickness measurement of 28 mm is above the typical range, further suggesting peripheral adiposity and insulin resistance. Diastolic blood pressure of 82 mmHg is mildly elevated. The diabetes pedigree function score of 0.52 indicates a moderate but significant hereditary predisposition to the condition. Serum insulin at 96 μ U/mL is within the normal range, though borderline, and should be monitored in the context of the elevated glucose.

Clinical Impression

The clinical profile of Sarah K. converges across multiple established risk dimensions for Type 2 Diabetes Mellitus: elevated fasting glucose, Class I obesity, increased skin thickness indicating peripheral insulin resistance, moderate hereditary risk, advanced maternal age, high parity, and mildly elevated blood pressure. The fasting glucose of 135 mg/dL alone meets the ADA diagnostic threshold for diabetes (≥ 126 mg/dL on confirmed testing). A formal confirmatory test (repeat fasting glucose or HbA1c) is required before a clinical diagnosis is established. This report does not constitute a standalone formal diagnosis.

Recommendations

- Confirmatory testing:** Repeat fasting plasma glucose on a separate day, or obtain HbA1c — a result $\geq 6.5\%$ confirms Type 2 Diabetes Mellitus per ADA criteria.
 - Oral Glucose Tolerance Test (OGTT) to assess postprandial glucose handling given the history of multiple pregnancies and potential prior gestational diabetes.
 - Initiate structured lifestyle intervention: caloric restriction targeting 5–7% weight reduction, and at least 150 minutes of moderate aerobic activity per week.
 - If diagnosis confirmed, initiate first-line pharmacotherapy (Metformin) per ADA guidelines.
 - Blood pressure monitoring and management — diastolic elevation in diabetic patients accelerates microvascular complications.
 - Lipid panel, renal function (eGFR, urine albumin), and ophthalmology referral if diabetes is confirmed, to establish baseline organ status.
 - Genetic counseling consideration given DPF of 0.52 and high parity history.
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Examining Clinician — Endocrinology

Date: April 29, 2026

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Laboratory findings do not constitute a standalone formal diagnosis.